


```

clc;clear;close all;
%import data
filepath=('C:\Users\PC\Desktop\MATLAB 1 ASS.xlsx');
Data=readtable(filepath,"VariableNamingRule","preserve");
disp(Data);

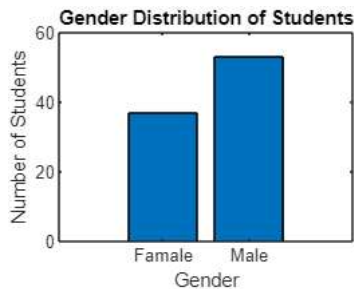
```

NAME	REGISTRATION NO	PROGRAM	TRIBE	ASSOCIATION
{ 'Akiteng Deborah' }	{ 'BU/UP/2025/1531' }	{ 'WAR' }	{ 'Iteso' }	{ 'TESDA' }
{ 'Akwir mercy Leah' }	{ 'BU/UP/2025/1532' }	{ 'WAR' }	{ 'Lango' }	{ 'NUSA' }
{ 'Auma Pretty' }	{ 'BU/UP/2025/2109' }	{ 'WAR' }	{ 'Lango' }	{ 'NUSA' }
{ 'Elepot Erasmus Wandera' }	{ 'BU/UP/2025/2113' }	{ 'WAR' }	{ 'Iteso' }	{ 'NUSA' }
{ 'Opus Jonah' }	{ 'BU/UP/2025/2142' }	{ 'WAR' }	{ 'Iteso' }	{ 'TESDA' }
{ 'Kinobe Solomon James' }	{ 'BU/UP/2025/3265' }	{ 'WAR' }	{ 'Basoga' }	{ 'BUBSA' }
{ 'Bwire Elvis' }	{ 'BU/UP/2025/2110' }	{ 'WAR' }	{ 'Samia' }	{ 'BUSSA' }
{ 'Acheng Gift Oliver' }	{ 'BU/UP/2025/5144' }	{ 'WAR' }	{ 'Lango' }	{ 'NUSA' }
{ 'Okou Lawyers' }	{ 'BU/UP/2025/1560' }	{ 'WAR' }	{ 'Iteso' }	{ 'TESDA' }
{ 'Olinga Jonathan' }	{ 'BU/UP/2025/1562' }	{ 'WAR' }	{ 'Iteso' }	{ 'TESDA' }
{ 'Ainomugisha Promise' }	{ 'BU/UP/2025/3215' }	{ 'WAR' }	{ 'Baganda' }	{ 'BANKOSO' }

```

%plots
Data.GENDER = categorical(Data.GENDER);
%Gender distributionR,
figure('Name','Gender Distribution');
gendercounts=countcats(Data.GENDER);
bar(gendercounts);
set(gca,'xTickLabel',categories(Data.GENDER));
title('Gender Distribution of Students');
ylabel('Number of Students');
xlabel('Gender');
saveas(gcf, 'Gender_Distribution.png');

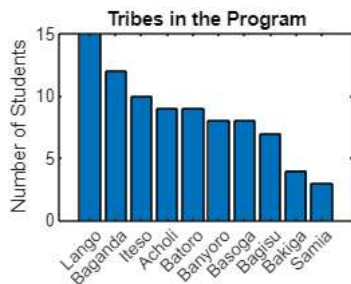
```



```

%Tribe distribution
figure('Name','Tribe Distribution');
tribecounts=groupcounts(Data,'TRIBE');
tribecounts=sortrows(tribecounts, 'GroupCount', 'descend');
numTribes = min(10, height(tribecounts));
bar(tribecounts.GroupCount(1:numTribes));
set(gca, 'XTickLabel', tribecounts.TRIBE(1:numTribes), 'XTickLabelRotation', 45);
title('Tribes in the Program');
ylabel('Number of Students');
saveas(gcf,'Tribe_Distribution.png');

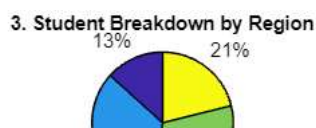
```



```

%STUDENTS BREAKDOWN
regionCounts = groupcounts(Data, 'REGION');
pie(regionCounts.GroupCount);
legend(cellstr(regionCounts.REGION), 'Location', 'southoutside', 'Orientation', 'horizontal');
title('3. Student Breakdown by Region');
saveas(gcf,'Students Breakdown by Region.png')

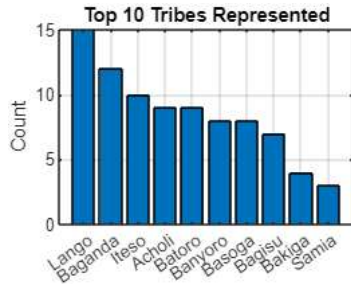
```



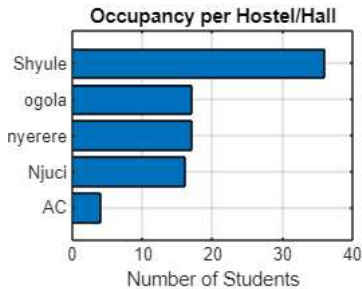


■ NORTHERN ■ WESTERN

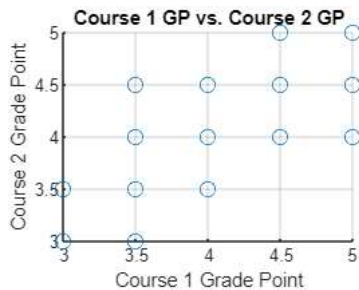
```
%Tribe distribution
tribeCounts = sortrows(groupcounts(Data, 'TRIBE'), 'GroupCount', 'descend');
numTribes = min(10, height(tribeCounts));
bar(tribeCounts.TRIBE(1:numTribes), tribeCounts.GroupCount(1:numTribes));
title('Top 10 Tribes Represented'); ylabel('Count');
xtickangle(35);
grid on;
```



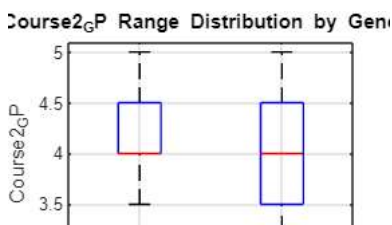
```
%Hostel/Halls
hostelCounts = sortrows(groupcounts(Data, 'HOSTEL/HALLS'), 'GroupCount', 'ascend');
barh(hostelCounts("HOSTEL/HALLS"), hostelCounts.GroupCount);
title('Occupancy per Hostel/Hall'); xlabel('Number of Students');
grid on;
saveas(gcf, 'Dashboard_Demographics.png');
```



```
%Academic Performance
scatter(Data.Course1_GP, Data.Course2_GP, 60);
xlabel('Course 1 Grade Point');
ylabel('Course 2 Grade Point');
title('Course 1 GP vs. Course 2 GP');
grid on;
```

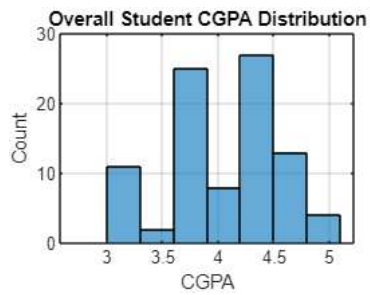


```
% GP by Gender
boxplot(Data.Course2_GP, Data.GENDER);
title('Course2_GP Range Distribution by Gender');
ylabel('Course2_GP');
grid on;
```

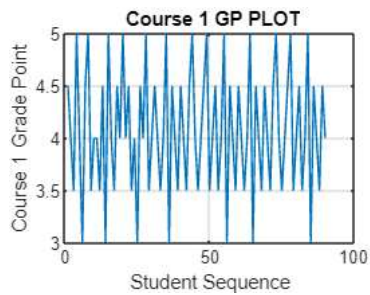




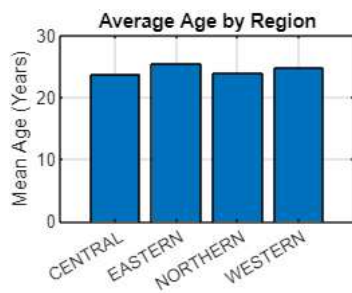
```
%CGPA Distribution
histogram(Data.CGPA);
title('Overall Student CGPA Distribution');
xlabel('CGPA');
ylabel('Count');
grid on;
```



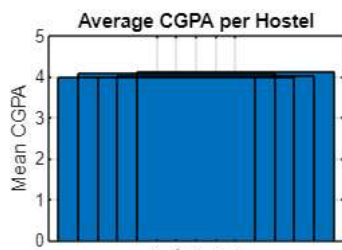
```
%Course_1GP LINE CHART
plot(Data.Course1_GP);
xlabel('Student Sequence');
ylabel('Course 1 Grade Point');
title('Course 1 GP PLOT');
grid on;
saveas(gcf, 'Dashboard_Geographic_Trends.png');
```



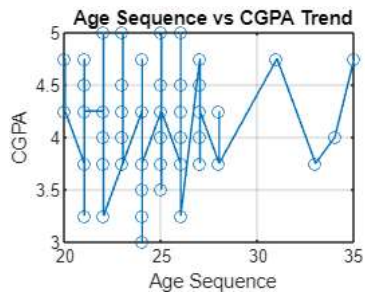
```
%AGE Distribution by Region
[rGroups, rNames] = findgroups(Data.REGION);
avgAgeRegion = splitapply(@mean, Data.AGE, rGroups);
bar(categorical(rNames), avgAgeRegion);
title('Average Age by Region');
ylabel('Mean Age (Years)');
grid on;
```



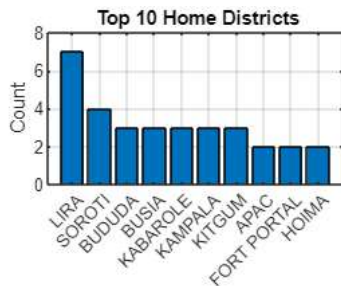
```
%Average CGPA Per Hostel
[hGroups, hNames] = findgroups(Data.("HOSTEL/HALLS"));
avgCGpaHostel = splitapply(@mean, Data.CGPA, hGroups);
bar(categorical(hNames), avgCGpaHostel, numTribes);
title('Average CGPA per Hostel');
ylabel('Mean CGPA');
grid on;
```



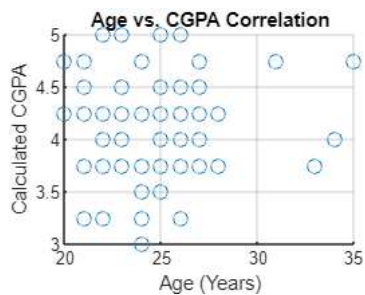
```
%CGPA Trend
[~, sortIdx] = sort(Data.AGE);
plot(Data.AGE(sortIdx), Data.CGPA(sortIdx), '-o');
xlabel(' Age Sequence'); ylabel('CGPA');
title(' Age Sequence vs CGPA Trend');
grid on;
```



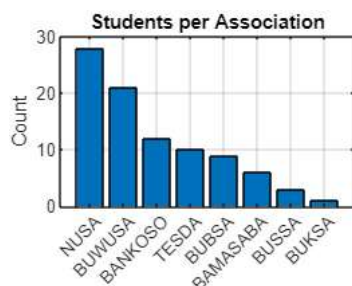
```
%TOP 10 HOME DISTRICTS
districtCounts = sortrows(groupcounts(Data, 'DISTRICT'), 'GroupCount', 'descend');
numDistricts = min(10, height(districtCounts));
bar(districtCounts.DISTRICT(1:numDistricts), districtCounts.GroupCount(1:numDistricts));
title(' Top 10 Home Districts');
ylabel('Count');
xtickangle(45);
grid on;
```



```
%AGE VS CGPA
scatter(Data.AGE, Data.CGPA, 50);
xlabel('Age (Years)');
ylabel('Calculated CGPA');
title('Age vs. CGPA Correlation');
grid on;
saveas(gcf, 'Dashboard_Academics.png');
```



```
assocCounts = sortrows(groupcounts(Data, 'ASSOCIATION'), 'GroupCount', 'descend');
bar(assocCounts.ASSOCIATION, assocCounts.GroupCount);
title('Students per Association'); ylabel('Count'); xtickangle(45); grid on;
assocCounts = sortrows(groupcounts(Data, 'ASSOCIATION'), 'GroupCount', 'descend');
```

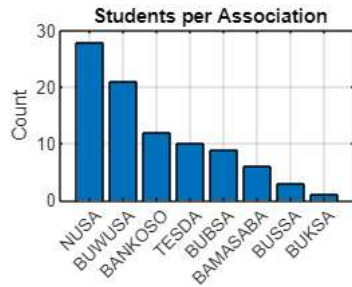


```
bar(assocCounts.ASSOCIATION, assocCounts.GroupCount);
```

```

title(' Students per Association');
ylabel('Count');
xtickangle(45);
grid on;

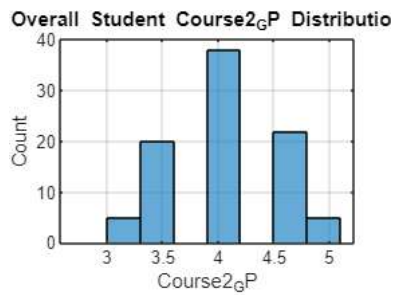
```



```

%Course2 GP distribution
histogram(Data.Course2_GP);
title('Overall Student Course2_GP Distribution');
xlabel('Course2_GP');
ylabel('Count');
grid on;

```



```

%STUDENT GENDER DISTRIBUTION
gCounts = groupcounts(Data, 'GENDER');
pie(gCounts.GroupCount, cellstr(gCounts.GENDER));
title(' Student Gender Split');

```

1. Student Gender Split

